WEEK 4 · DAY 1

Scoring a Detector + Quiz 2

The answer key comes out.
Every alarm from last week gets a report card. 🎯



NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

Rules first. Then the report cards.



Open the key

6 real events, finally revealed



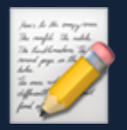
Score every alarm

recall, precision, false alarms



Count the cost

false alarms → engineer hours



Quiz 2

all of Week 3

⏱️ 10 min hook · 🎤 25 min slides · 👩🏾‍💻 50 min hands-on · 📝 25 min quiz + marking · 🏁 10 min wrap-up

QUICK QUESTION

Which mai-guard do you hire? 💂🏾



Musa shouts “Thief!” at every goat

10 thieves came this year. Musa caught **9**. He also shouted **200 times**. The estate no longer wakes up.

catches almost everything

wakes the whole estate



Ade sleeps all night

Ade never shouts. On **355 of 365** nights there was no thief, so Ade was “right” **97% of the time**.

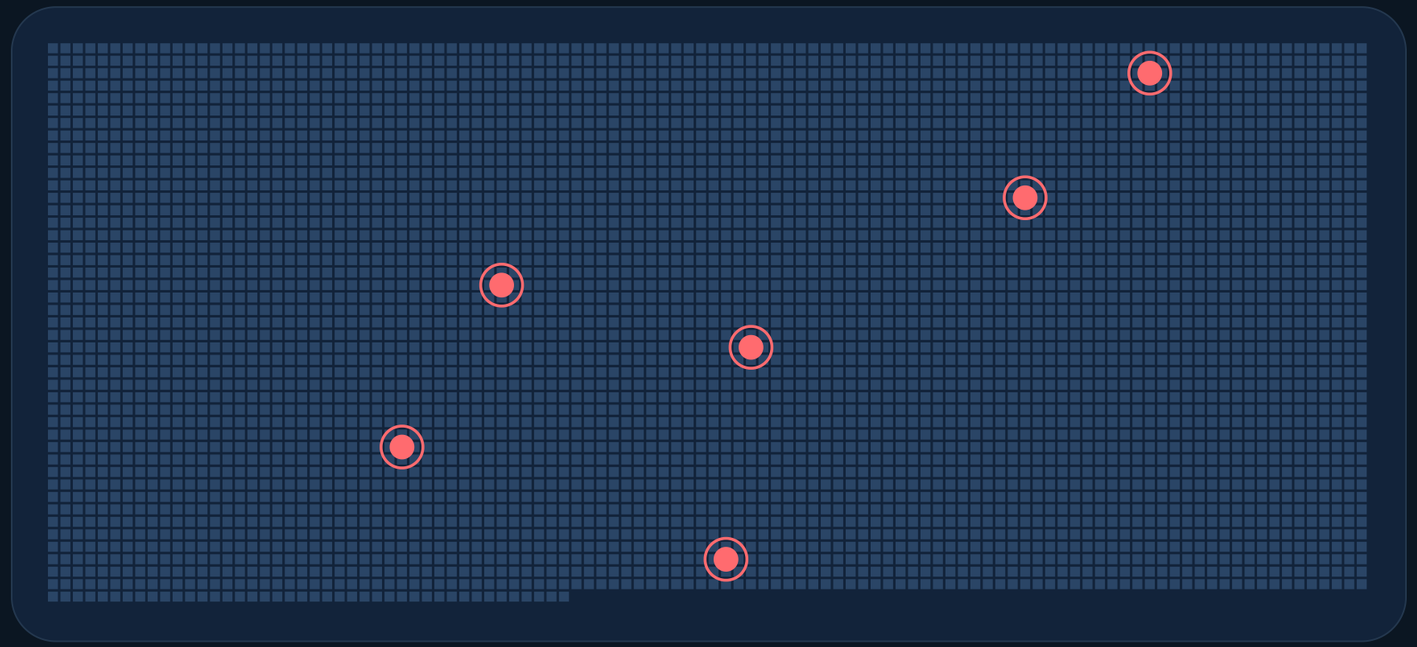
“97% accurate”

caught zero thieves

Neither. Today you learn the two numbers that expose both of them. 🎯

WHY “ACCURACY” LIES

A detector that never fires scores 99.87%



4,706

readings in last week's three files

6

real events hiding inside them

Say “no problem” every time and you're right 4,700 times out of 4,706. **Useless.**

PART 1

The answer key 🔑

For three days it was hidden. Now we open it.

TELEMETRY_ANOMALY_LABELS.CSV · 6 ROWS

Six real events, three files



3 spikes in bus power

1 spike + 1 level shift in the amplifier

1 slow drift in the battery

EV05 lasts 233 days · EV06 lasts 275

WRITE THE RULES BEFORE YOU COMPUTE ANYTHING

Four scoring rules

1 One flag inside = caught

A flag anywhere inside the event's start→end (±1 step) catches it.

2 Many flags = still ONE catch

EV02 flagged on both of its hours is one catch, not two.

3 Outside every event = false alarm

Each one is a page to the duty engineer at 2 a.m. 📟

4 A 233-day event is ONE event

Flag one day of EV05 and you caught it once, not 233 times.

Decide the rules first, so nobody can bend them after seeing the score. 🤦🏾

THE TWO NUMBERS THAT MATTER

Recall and precision



Recall

Of the **real events**, how many did we catch?

```
recall = caught / real_events
```

Musa: 9 ÷ 10 = **0.90** 👍🏾



Precision

When it **shouts**, how often is something really there?

```
precision = caught / all_alarms
```

Musa: 9 ÷ 200 = **0.045** 😩

Ade the sleeper: recall **0**. Precision? He never shouted, so there's nothing to measure.

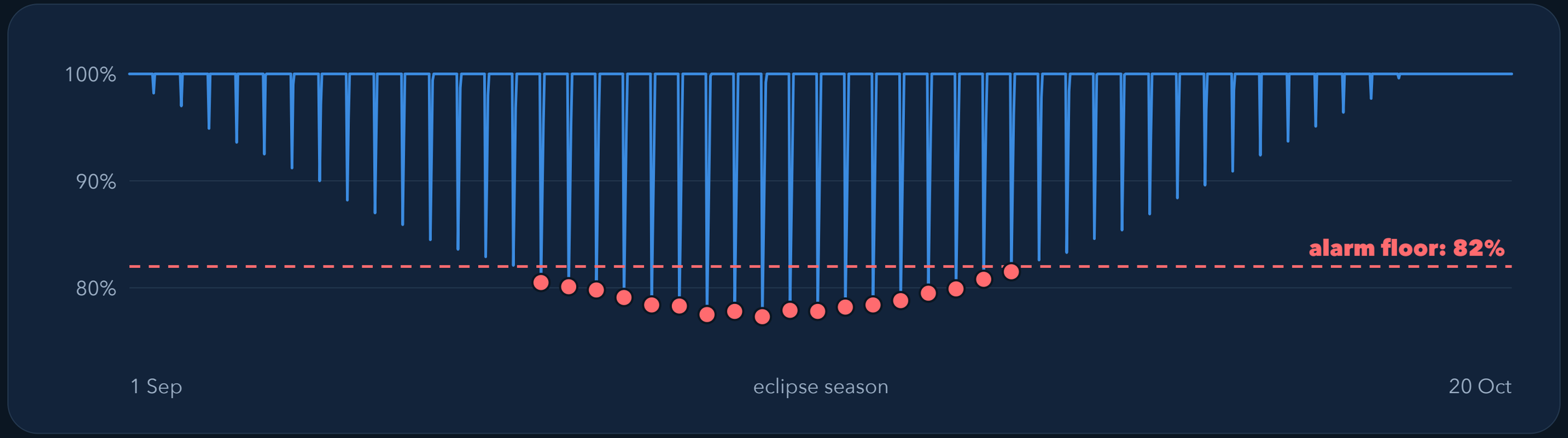
PART 2

Report cards 🎯

Same scorer. Same rules. Every detector from last week.

DETECTOR 1 · THE ALARM SOFTWARE'S FIXED FLOOR

State of charge < 82%: 18 alarms, 0 real

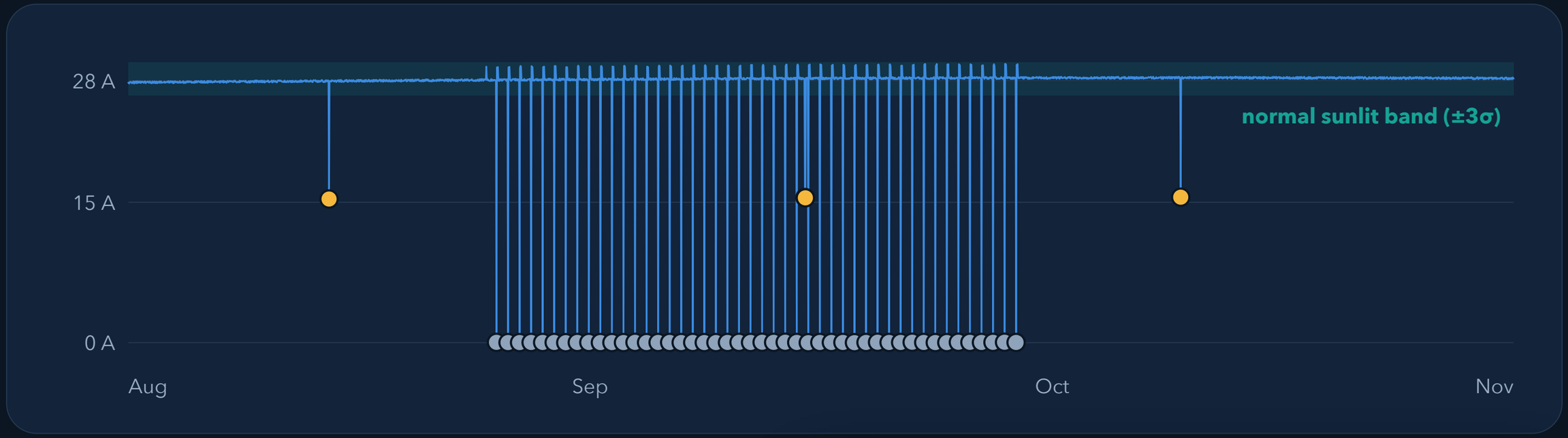


recall 0.00 **precision 0.00** **every alarm = a normal eclipse night**

real faults were in array current, not here

DETECTOR 2 · 3Σ CONTROL CHART ON SOLAR-ARRAY CURRENT

Same chart, one condition, a different life



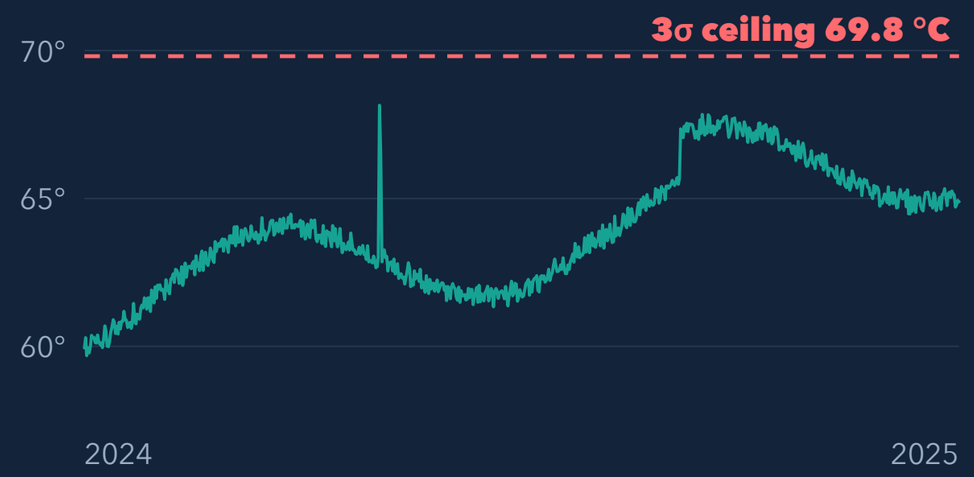
All hours: caught **3/3** · **46 false alarms**
grey dots, every one at 21:00 UTC (eclipse) · precision 0.06

Sunlit hours only: caught **3/3** · **0 false alarms**
gold dots = the real dropouts · precision **1.00** ✅

DETECTORS 3 + 4 · THE AMPLIFIER (TWTA)

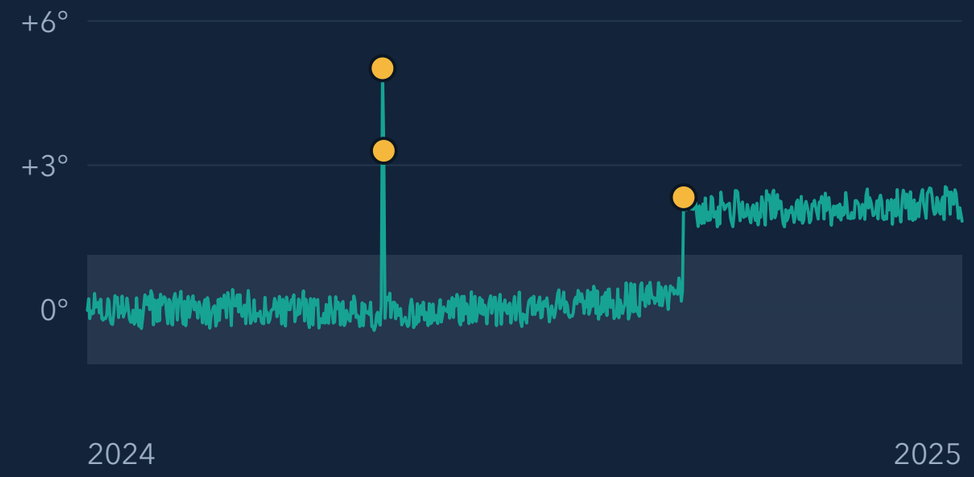
Raw values hide it. The residual shouts.

🌡️ Raw temperature, 3σ ceiling



Highest day 68.2 °C never reaches 69.8 °C → caught **0/2**

🧮 Residual = actual − expected



Spike = 13σ, shift stays above the band → caught **2/2**, **0** false alarms

DETECTOR 5 · CUSUM ON THE SAME RESIDUAL

More sensitive is not free

2/2

events caught (same as the residual rule)

2

false alarms: 21 Apr, 06 May

0.50

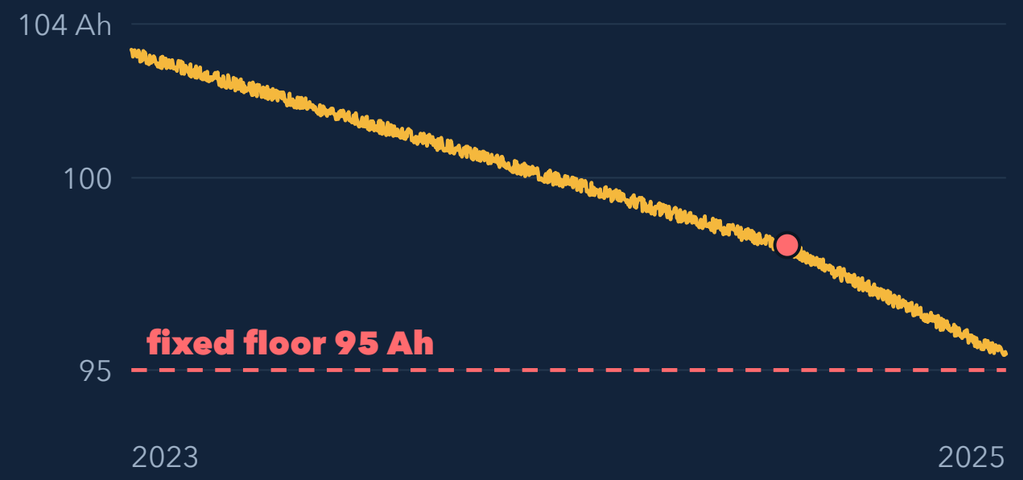
precision (residual rule: 1.00)

CUSUM adds up small pushes, so it also adds up small *innocent* pushes.
The scorer is how you find out which one wins **on your data**.

DETECTOR 6 · THE BATTERY

The fault no value-alarm can see 🪫

Capacity vs a fixed 95 Ah floor



Lowest day 95.4 Ah: never below 95 → caught **0/1**

🌶️ 90-day fade rate (the slope)



Rate alarm catches EV06 **31 days** after the knee, **0** false alarms

THE REPORT CARD

One scorer. Seven detectors.

	DETECTOR	CAUGHT	FALSE ALARMS	RECALL	PRECISION
🔋	**SoC floor < 82%** · bus power · W3D1	0/3	18	0.00	0.00
📊	**Control chart, all hours** · bus power · W3D2	3/3	46	1.00	0.06
🌞	**Control chart, sunlit only** · bus power · W3D2	3/3	0	1.00	1.00
🌡️	**Raw 3σ on temperature** · TWTA · W3D2	0/2	0	0.00	—
🧱	**Residual 3σ (drive + season + age)** · TWTA · W3D2	2/2	0	1.00	1.00
➕	**CUSUM on the residual** · TWTA · W3D3	2/2	2	1.00	0.50
🪫	**Fixed floor 95 Ah** · battery · W3D3	0/1	0	0.00	—

PART 3

Count the cost 

Managers don't decide on “precision 0.06”. They decide on hours.

FALSE ALARMS → ENGINEER HOURS

The all-hours chart costs 31 hours a year

46
false alarms per eclipse season

→

× 2
eclipse seasons a year

→

× 20
minutes to check each one (assumption)

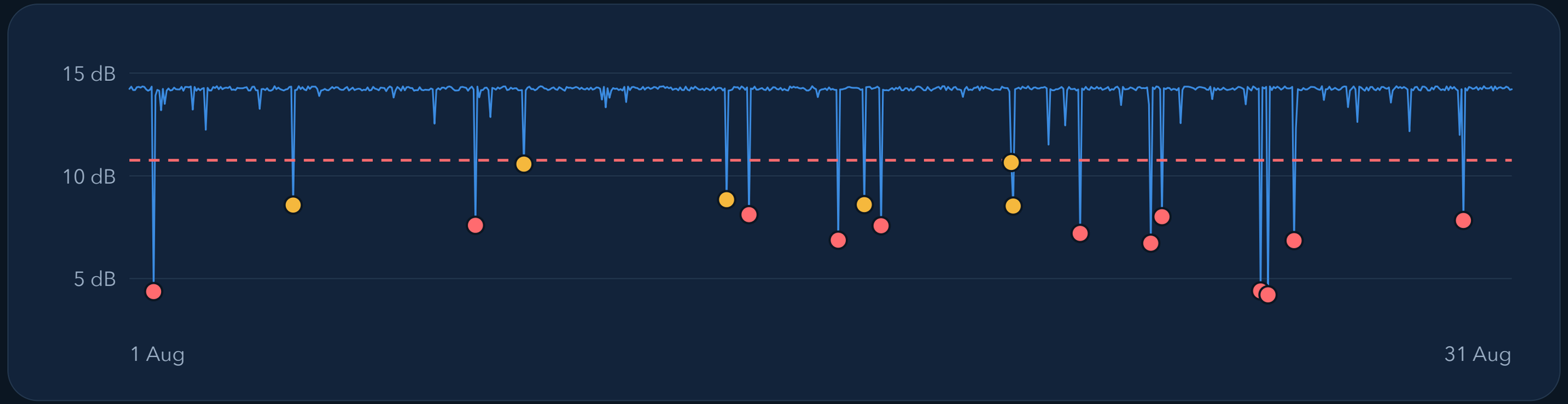
→

31 h
≈ 4 full 8-hour shifts

"Add one condition and get 4 shifts back" is a decision a manager can make. 🤦🏾‍♀️

THE RF FILE · AUGUST, HOURLY

Unusual is not the same as anomaly 🌧️



18 flagged hours · 🔴 **12 real outages** · 🟡 **6 rain, link held** · **precision 0.67**

dashed line = naive 3σ (10.8 dB)

WHICH NUMBER LEADS?

It depends on what a miss costs



Recall first

Battery, power, propulsion. A miss can cost the satellite.
NigComSat-1 was lost in 2008 after a solar-array failure. Accept some extra pages.



Precision first

Rain fade, NOC dashboards, anything that pages people all night.
Too many false alarms and engineers **stop reading the pager**. That's alarm fatigue.

Write down which one leads **before** you score, and say why. ✍🏾

QUIZ 2 · WEEK 3: ANOMALY DETECTION

30 questions. 20 minutes. Just you. 

Phones down

notebooks closed, no AI

One letter

circle A, B, C or D (or True/False)

No “always B”

answers are spread across A–D

Mark together

10 minutes, swap and tick

Covers alarms (W3D1) · control charts + residuals (W3D2) · drift + CUSUM (W3D3) · scoring (today).

YOUR TURN · 50 MINUTES

Be the referee 👩🏾‍⚖️

1. Open **W4D1-exercise**, save a copy, run STEP 0
2. **A** Rules first: accuracy trap, Musa's numbers
3. **B** Score 6 detectors with one scorer
4. **C** Turn false alarms into hours; the rain lesson



14 checks to pass

🌶️ Pepper task: build the slope alarm that finally catches EV06

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know



Accuracy lies when events are rare. Never report it for a detector.



Recall = did we miss anything? **Precision** = when it pages, is it real? Rules written **first**.



False alarms cost hours. Say it in hours and managers can decide.

Next time 👉 **Meet scikit-learn**: faults hiding between two normal-looking sensors. 🔍